Introduction

This is a document outlining the workflow of data preprocessing for the task of fine-tuning Prithvi-2.0-EO (crop map classification). This example is using a single sentinel-2 40 band raster tile, and the land cover crop map vector data. It's assumed the user has basic QGIS skills (not overly detailed about loading data etc).

Preprocessing in QGIS

Sentinel-2 Tile

The goal here is to rearrange and reduce the bands from the original 40 to the 18 that Prithvi-2.0-EO requires. Using the **Rearrange bands** tool, select the bands shown in Figure 1. Press run and inspect the bands afterwards to ensure they're in the correct order.

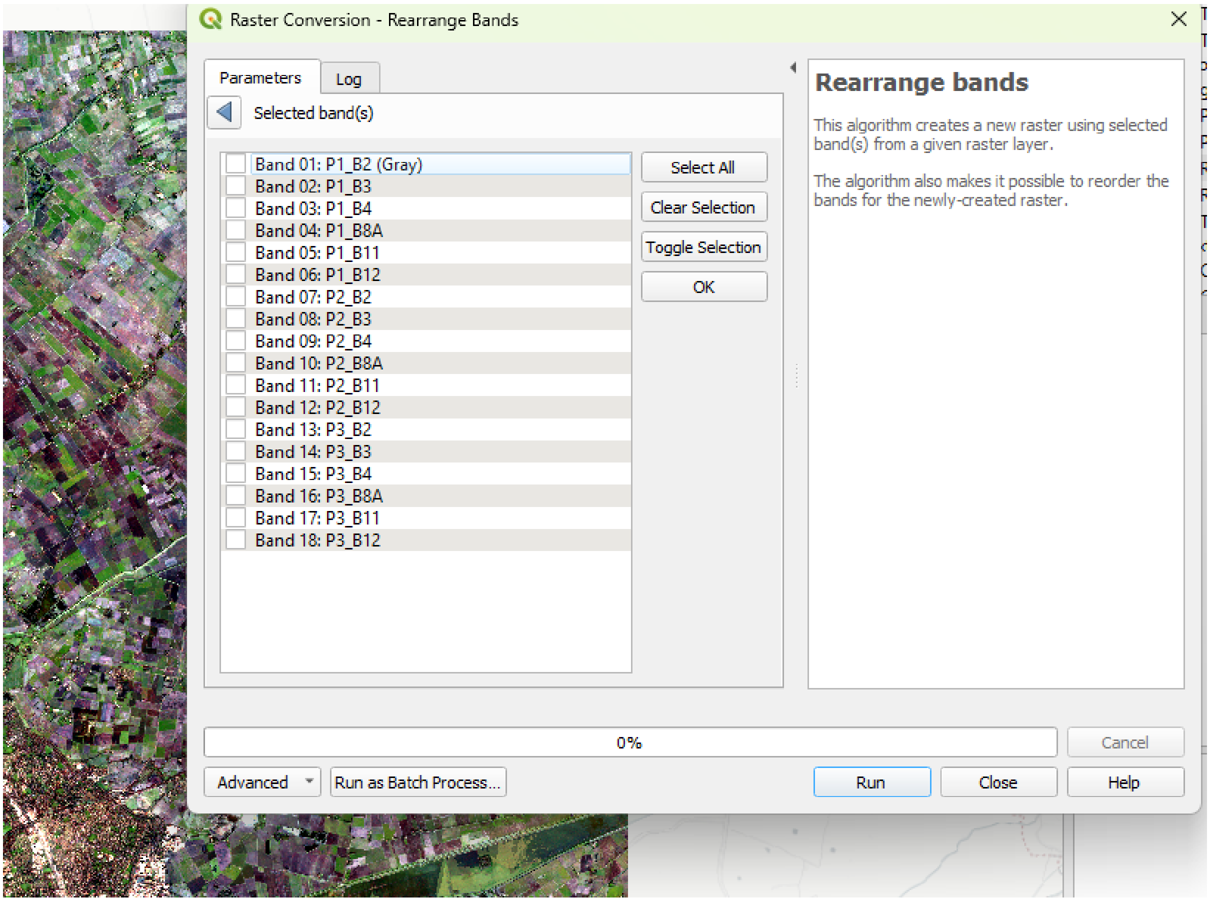

Figure 1: Band Order.

Vector Crop Map Data

The task here is to reclassify the vector data and rasterize it. For the vector crop map data, you want to first reclassify the crop name attributes (strings) into classes (integers). Open the attribute table and use the **field calculator** to create a new field titled "class", and in the expression box paste:

```
CASE WHEN "crop2024" = 'be' THEN 1 WHEN "crop2024" = 'fs' THEN 2 WHEN "crop2024" = 'fw' THEN 3 WHEN "crop2024" = 'gr' THEN 4 WHEN "crop2024" = 'ma' THEN 5 WHEN "crop2024" = 'or' THEN 6 WHEN "crop2024" = 'ot' THEN 7 WHEN "crop2024" = 'pe' THEN 8 WHEN "crop2024" = 'po' THEN 9 WHEN "crop2024" = 'sb' THEN 10 WHEN "crop2024" = 'sl' THEN 11 WHEN "crop2024" = 'so' THEN 12 WHEN "crop2024" = 'sw' THEN 13 WHEN "crop2024" = 'wb' THEN 14 WHEN "crop2024" = 'wo' THEN 15 WHEN "crop2024" = 'ww' THEN 16 ELSE 0 END
```

Upon confirming this your attribute table should now look like Figure 2.

lcp_2024_final — Features Total: 1790698, Filtered: 1790698, Selected: 0

	gid	poly_id	parent	crop2024	new2023	gid_lcm	wc24_raw	wc24_class	class
1	1993367	986796	0	sb	0	6525240	0.219550654292...	S1	10
2	1536581	542886	0	gr	0	5336478	0.460507363080...	NULL	4
3	1032925	2632672	0	sb	0	5265458	0.1825110912323	S1	10
4	2763995	2909236	0	sb	0	4502531	0.467977821826...	S2	10
5	271616	2020466	0	or	0	1651507	0.673469364643...	NULL	6
6	2561945	3167766	0	gr	0	4266537	0.805233716964...	NULL	4
7	298542	2896038	0	sb	0	3896239	0.586223602294...	S2	10
8	5657069	2697841	1240304	sb	0	5582337	0.273224979639...	S2	10
9	1695014	5201948	0	sb	0	688709	0.550592094659...	S2	10
10	401775	2231844	0	gr	0	18871	0.690697103738...	NULL	4
11	1532389	2906770	0	sb	0	5530375	0.159551665186...	S1	10
12	2619647	1192224	0	gr	0	4337847	0.35464346408844	NULL	4
13	2064489	4262529	0	or	0	5879204	0.773779988288...	NULL	6
14	1168496	2825484	0	ot	0	1918248	0.576512455940...	NULL	7
15	1970566	2881840	0	gr	0	1101371	0.261995017528...	NULL	4
16	2797042	6750156	NULL	gr	0	4027845	0.353887408971...	NULL	4
17	917980	1353214	0	gr	0	5709389	0.618062645196...	NULL	4
18	70136379	NULL	NULL	gr	1	6368096	0.678049296140...	NULL	4
19	643920	32624	0	or	0	36753	0.757905304431...	NULL	6
20	548598	0	34395	or	0	5281558	0.771838545799...	NULL	6
21	1390522	1377820	0	gr	0	2835416	0.757254809141...	NULL	4
22	906795	375148	0	or	0	1407526	0.548800736665...	NULL	6
23	1172912	910885	0	gr	0	4573432	0.739478945732...	NULL	4

Figure 2: Attribute table with Class Field added.

Now we will rasterize the vector data. Open the **Vector Conversion – Rasterize (Vector to Raster)** tool and use the crop map vector layer as the input layer. For the field to use for a burn-in value select the “class” field we just created. For output raster size units select “Georeferenced units”, with the Width/Horizontal and Height/Vertical resolution both set to 10. Finally set the output extent to your sentinel-2 tile so the pixels align properly. See Figure 3 for parameters and run.

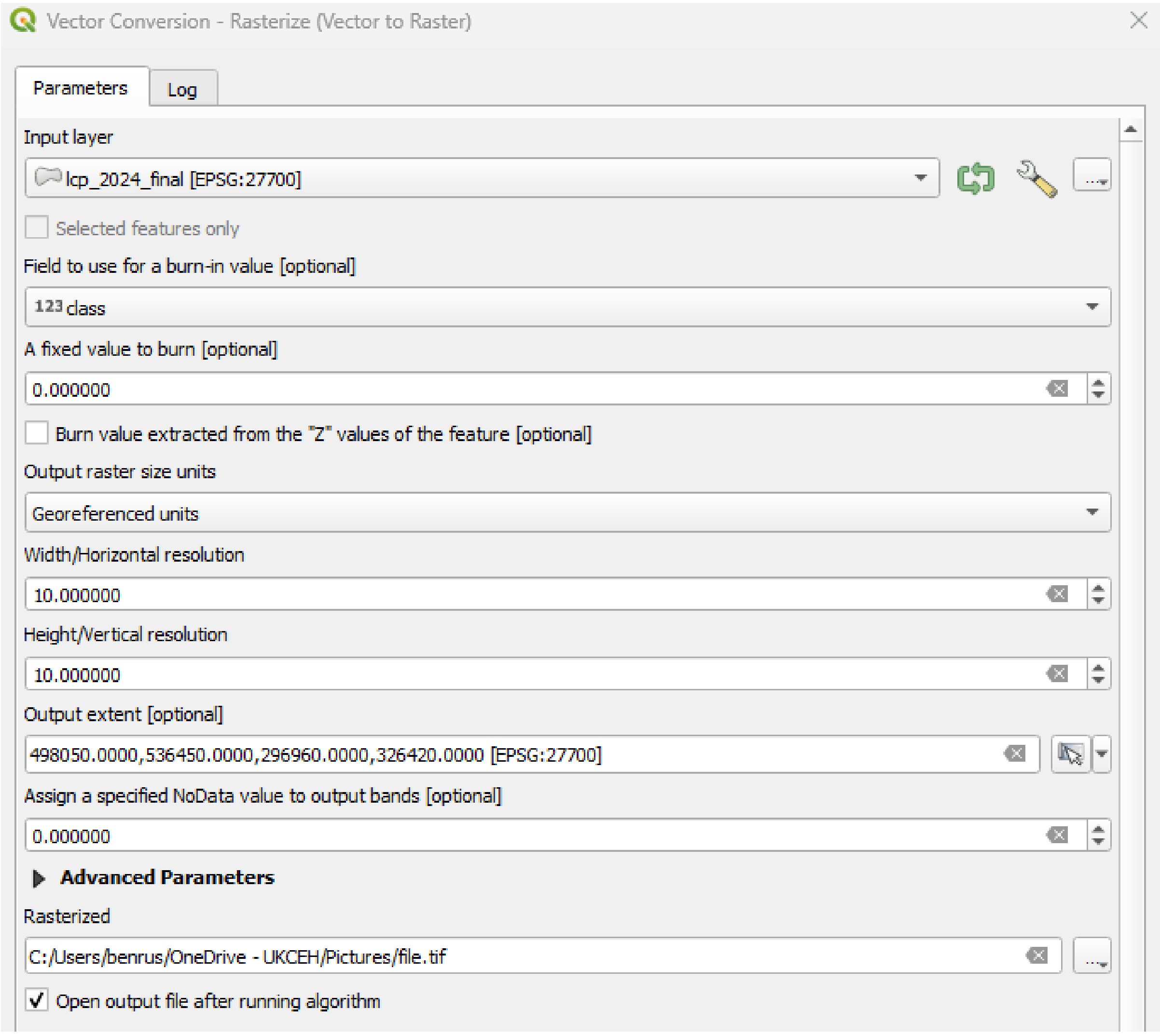

Figure 3: Rasterize parameters.

To ensure the final output has been created correctly, open the rasterized layer's symbology and select the Palleted/Unique Values render type. Classify it with 16 classes and a random colour ramp. After applying, the rasterized layer should now look something like Figure 4.

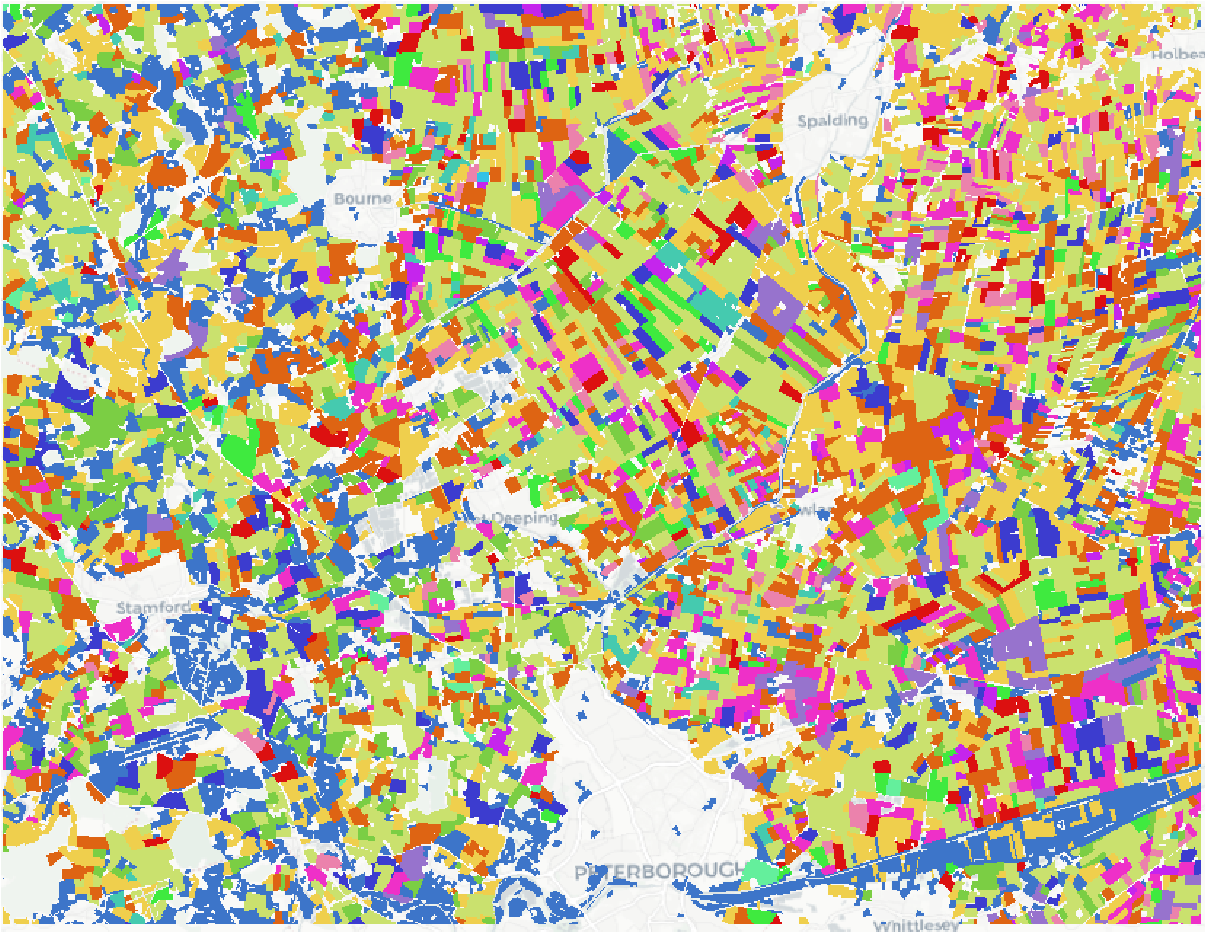

Figure 4: Rasterized Layer.

Data Extraction & Chip Generation (Grid)

We now need to create a grid that will act as our mask extraction layer. Open the **Create Grid** tool, set the grid type to rectangle, grid extent to the sentinel-2 tile, Horizontal/Vertical spacing as 2,240 metres, and the grid CRS to EPSG:27700. See Figure 5 for the parameters and then run.

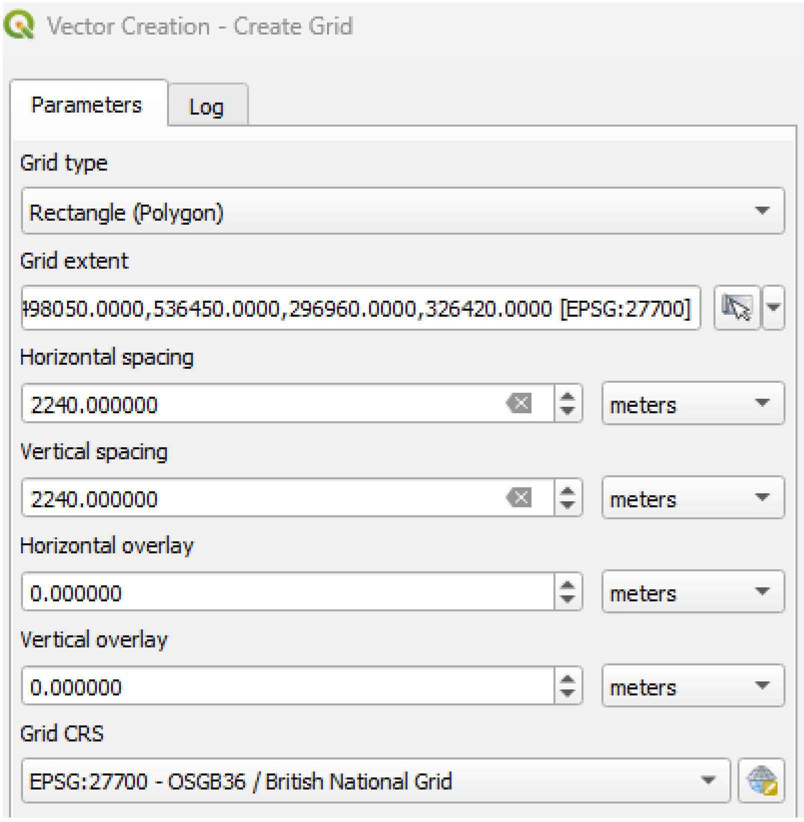

Figure 5: Grid creation parameters.

Now we will generate chips using the grid mask. Open the **Raster Extraction – Clip Raster by Mask Layer** tool and set the input layer as the sentinel-2 raster tile. Set the mask layer as the grid and select the green arrows that "iterate" over the layer. Set the source and target CRS as EPSG:27700, and target extent as the sentinel-2 tile. Untick "open output file after running algorithm" and leave the rest of the parameters. Save your chips in a folder named "images". Run, and keeping the same parameters change the input layer to the rasterized layer, saving the chips in another folder named "masks". See Figure 6 for the parameter settings.

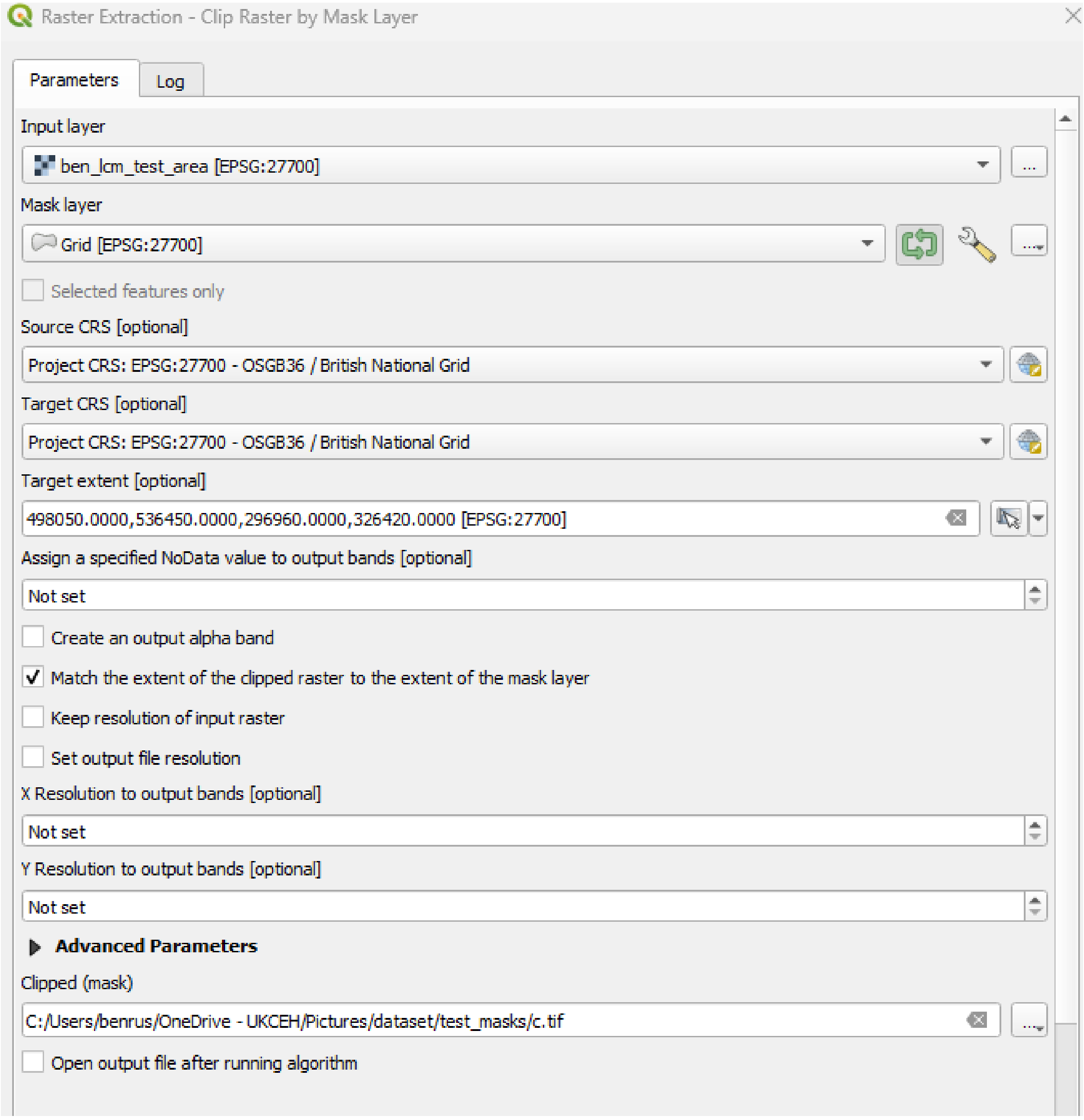

Figure 6: Extraction parameters.

Conclusion

By following this workflow, you should now have the images and mask labels in separate folders formatted for Prithvi-2.0-EO input. Whatever you named the chips, they should all end with a number, it is important these numbers are matching between the two sets of chips.